

Data Structures Lab

Lab Experiment: 11

Objective:

To implement a graph traversal technique (BSF/DFS) in C.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define MAX 100 // Maximum number of vertices
```

```
void bfs(int adjacencyMatrix[MAX][MAX], int visited[MAX], int numVertices, int startVertex) {
```

```
    int queue[MAX], front = -1, rear = -1; // Queue for BFS
```

```
    int i;
```

```
    // Mark the start vertex as visited and enqueue it
```

```
    visited[startVertex] = 1;
```

```
    queue[++rear] = startVertex;
```

```
    printf("BFS Traversal: ");
```

```
    while (front != rear) { // While the queue is not empty
```

```
        // Dequeue a vertex
```

```
        int currentVertex = queue[++front];
```

```
        printf("%d ", currentVertex);
```

```
        // Check all adjacent vertices
```

```
        for (i = 0; i < numVertices; i++) {
```

```

        if (adjacencyMatrix[currentVertex][i] == 1 && visited[i] == 0) {
            // If the vertex is adjacent and not visited, enqueue it
            queue[++rear] = i;
            visited[i] = 1;
        }
    }
}
printf("\n");
}

```

```

int main() {
    int numVertices, numEdges;

    int adjacencyMatrix[MAX][MAX] = {0}; // Initialize adjacency matrix with 0s
    int visited[MAX] = {0}; // Visited array initialized to 0

    printf("Enter the number of vertices: ");
    scanf("%d", &numVertices);

    printf("Enter the number of edges: ");
    scanf("%d", &numEdges);

    printf("Enter the edges (u v) where u and v are vertex indices (0 to %d):\n", numVertices - 1);
    for (int i = 0; i < numEdges; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        adjacencyMatrix[u][v] = 1; // Mark edge u->v
        adjacencyMatrix[v][u] = 1; // Mark edge v->u for undirected graph
    }
}

```

```
int startVertex;  
  
printf("Enter the starting vertex for BFS: ");  
scanf("%d", &startVertex);  
  
// Perform BFS  
bfs(adjacencyMatrix, visited, numVertices, startVertex);  
  
return 0;  
}
```

```
Enter the number of vertices: 4  
Enter the number of edges: 3  
Enter the edges (u v) where u and v are vertex indices (0 to 3):  
1 0  
0 2  
2 3  
Enter the starting vertex for BFS: 2  
BFS Traversal: 2 0 3 1  
  
...Program finished with exit code 0  
Press ENTER to exit console. 
```